Lösungen · Trigonometrie

1. $\boldsymbol{\varepsilon = \cos^{-1}(\frac{13}{32}) \approx 66{,}0°}$ ..... $\cos\varepsilon = \frac{13}{32}$

2. $\boldsymbol{\beta_1 = \cos^{-1}(\frac{255}{384}) \approx 48{,}4°}$ ..... $\cos\beta_1 = \frac{255}{384}$

3. $\boldsymbol{\alpha = \sin^{-1}(\frac{7{,}4}{14{,}1}) \approx 31{,}7°}$ ..... $\sin\alpha = \frac{7{,}4}{14{,}1}$

4. $\boldsymbol{x = 2\sqrt{7289} \approx 170{,}8\text{ cm};\ \alpha = \tan^{-1}(\frac{16}{170}) \approx 5{,}4°}$ ...
$x^2 = 170^2 + 16^2$; $\tan\alpha = \frac{16}{170}$

5. $\boldsymbol{\alpha = \cos^{-1}(\frac{4}{9}) \approx 63{,}6°}$ ..... $\cos\alpha = \frac{4}{9}$

6. $\boldsymbol{\alpha = \tan^{-1}(\frac{14}{9}) \approx 57{,}3°}$ ..... $\tan\alpha = \frac{14}{9}$

7. $\boldsymbol{BC = \sqrt{20} \approx 4{,}47;\ \beta = \tan^{-1}(2) \approx 63{,}4°}$ ... $BC^2 = 2^2 + 4^2$; $\tan\beta = \frac{4}{2}$